Disobedience in normative multi-agent systems

Anonymous Author(s)
Submission Id: 1370

ABSTRACT

When an autonomous agent deliberates whether to disobey, it must anticipate how its behaviour will be interpreted and acted upon by the surrounding system. In this paper, we distinguish four types of disobedience: intentional violation, justified exception, civil disobedience, and trolling. Monitoring must therefore go beyond the simple question of whether a violation occurred, and instead determine what kind of disobedience is at issue. While violations and exceptions are well studied, civil disobedience and trolling introduce conceptually new categories of behaviour. Each type requires a distinct enforcement workflow: intentional violations are sanctioned, exceptions are waived, civil disobedience is sanctioned but also logged as a reform signal, and trolls are sanctioned but excluded from reform processes. To capture this, we formalize a compliance management architecture that separates monitoring of observable behaviour, assessment of disobedience type, and dispatching to the appropriate enforcement workflow. This separation clarifies the dual perspective: agents use reason-based practical reasoning to decide whether to obey or disobey, while the governance framework processes observable outcomes and routes them into differentiated institutional responses.

ACM Reference Format:
Anonymous Author(s). 2026. Disobedience in normative multi-agent systems. In *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026), Paphos, Cyprus, May 25 – 29, 2026,* IFAAMAS, 9 pages.

INTRODUCTION

Multi-agent systems (MAS) are real-world or digital environments in which different agents, human and artificial, pursue their goals. To ensure conviviality and efficiency in the MAS, it is typical to use *norms* to guide the behaviour of agents [6, 11]. Special agents or frameworks that are tasked with the role of instituting and maintaining norms are called *institutions* [20].

Norms are an instrument for incentivising behaviour for intelligent and autonomous agents. Computational artefacts that do not have agency can be managed by constraints. Unlike constraints, norms can be violated and they “tolerate" exceptions. That is, it is not the case that when the conditions are satisfied, the deontic statement holds. For the deontic statement to hold, it needs to be *detached* from the norm. The decision to detach is the task of *the institution.* Institutions monitor *violation of the norms*, remove deontic statements, and impose sanctions where appropriate.

Being able to reason about norms is one of the core competences of autonomous agents. This means not only being able to act while satisfying norms, but also being able to violate them. Norm violations can occur under different circumstances and can be unintentional and intentional. Unintentional violations occur when the agent did not know that there is a norm that needs to be considered. We are here concerned with intentional norm violations. We refer to intentional norm violations as *disobedience.*

It is important for artificial agents to be able to disobey a norm. Only in small controlled environments, it is possible to exhaustively prescribe rules and constraints that exactly cover every desirable or undesirable situation that can arise. There are some situations in which acting rationally and responsibly means disobeying a norm [3, 30, 31]. Furthermore, emergent properties, pluralism, and drift make ability for non-compliance necessary in a bounded reality.

Norm monitoring is the process of identifying norm violations in a MAS [9, 15]. Recently, He et al. [23] explored the possibility of using large language models (LLMs) to detect violations of norms. *We propose that it is important not only to discern between intentional and unintentional norm violation (disobedience) but also to identify the reasons for disobedience.* Disobedience management complements ethics-by-design [8, 17] by specifying how to classify, justify, and sanction/waive violations.

Multi-agent systems are not only systems of artificial agents. In a digital environment, the artificial agents can act as the norm monitoring institutions of the environment shared with people. It is easier to constrain people activities in a digital environment compared to the same activity in the physical world. That brings about the temptation to use constraints to replace norms. This is problematic because by eliminating the possibility that people can violate norms, we can violate some of the basic rights of people and decrease the safety in society. Norm violators and disobedient agents can be a symptom of a poorly functioning normative system that does not serve the purpose it is constructed for. If agents unintentionally violate norms, then this is a symptom of lack of transparency about what normative system is in place or to whom it applies. If agents, human or artificial, disobey, the reasons for their disobedience can point to the need for improvement in the norm management.

Having well-managed intelligent multi-agent systems is becoming particularly important with the increased capability of Agentic AI [7]. In this paper, we focus on two problems. The first is the problem of distinguishing between violations and disobedience, but also between different types of disobedience. Sufficient, and we would say mathematical, understanding of the epistemology is necessary to address the second problem. The second problem is what abilities do agentic AI and need to have in order to disobey and how can an institution recognise the nature of their norm violation to a degree that the appropriate MAS institutional measures are taken to address it.

Methodology. We use an analytical approach. We consider motivating examples, and where possible the existing literature, to identify the different types of how an agent can disobey. We then

Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026), C. Amato, L. Dennis, V. Mascardi, J. Thangarajah (eds.), May 25 – 29, 2026, Paphos, Cyprus.

consider the institutional view and design an architecture for disobedience classification and response. We lastly design a reasoning approach for agents to reason about violating a norm.

Scope. We focus on an environment in which there is a single normative system. We assume agents that are autonomous, intelligent and rational in the standard sense of [33], but for whom no internal architecture is known. The agent is accountable to the institution; that is, the institution can impose norms that the agent is expected to abide by, and the institution can sanction the agent.

EXAMPLES OF INTENTIONAL VIOLATIONS

Intuitively a norm is a rule that prescribes what is obligatory, permissible, impermissible (prohibited), omissible in a given situation. To obey a norm means to act in accordance to what the norm prescribes in the specified situation, while to violate it norm means to act contrary to what the norm prescribes. To elucidate the differences between the differences in disobedience, we consider some motivating examples. We consider the related literature and then use it to analyse the examples for insights to specify requirements for reasoning abilities vis disobedience for institutions and agents in MAS.

Example 1. The agent is a driverless car taxi that is driving in a 50km/h speed limitation zone. There is no other cars on the road and the conditions are dry with good visibility. It speeds to get to the destination faster. There is a financial fine for speeding.

Example 2. A house robot is asked to bring a cold beer from the fridge to its owner. The norm is to obey the requests of the human. The human has already been served two beers, and the battery of the robot is low. The Robot decides to go to the charging station instead. The human may be too drunk to notice. The sanction is a complaint to the manufacturer and a claim for reimbursement. [1]

Example 3. The agent is a driverless car taxi that is driving in a 50km/h speed limit zone. The passenger is in a medical emergency. The car speeds to get to the destination faster. There is an expectation that there will be no fine because there is a good cause for the violation.

Example 4. A robot guide dog should obey the commands of the human it serves. The user gives him the command to go forward because the pedestrian crossing light is "green". The dog moves in the opposite direction because it detects an incoming car running a red light. The dog has assessed that it was unsafe to follow the command[2].

Example 5. An intelligent content moderator operates on a social media platform that does not allow the posting of explicit content. The norm is that the content should be suitable for work and PG 13. Female nipples count as explicit content. Users who post explicit content are banned. However, public health is very important. To improve breast cancer treatment, early detection is crucial and to ensure early detection tutorials on how to do self-exam are posted and shared online. Public acceptance of breastfeeding improves the health of babies and the safety of mothers. Images of public breastfeeding shared on social networks contribute to normalising this aspect of human physiology that is taboo in many contexts. Activists have taken to posting images of female nipples, despite being banned when they do, to promote the requirement that this imagery should not count as explicit content. Other users can see that some activists are banned and find out why on other social media. The social media platform can start losing revenue from advertisers when the expectation that activists should be unbanned reaches critical momentum.

Example 6. A driverless car always drives above the speed limit, whatever that is. The owner does not care about punishments. They want the world to know that they are rich enough to afford the law not applying to them.

[1] Based on an example from [12].

[2] The example is based on an example from [31].

RELATED WORK.

A disobedience is not a dilemma. Let us start by observing that disobeying is not a moral dilemma. Cristani and Burato [14] define a moral dilemma as a question regarding two actions $act1$ and $act2$, which are 'morally equivalent'[3], and an agent: What should the agent do between $act1$ and $act2$? Their definition can be applied to any normative dilemma, if the 'morally equivalent' can be replaced with 'equivalently violating the same norm'. When an agent considers violating the norm $act1$, or not violating it $act2$, clearly $act1$ and $act2$ are not normatively equivalent.

We start with the literature on norm violation.

Norm monitoring. The literature of MAS is mainly concerned with norm monitoring, that is, detecting and sanctioning violations; see, for example, [3, 9], than with the permissibility and desirability of disobedience. Norm violation and thus disobedience is discussed more in moral philosophy. Consequential moral theories by definition would consider disobedience to be moral when it is in the interest of attaining the right consequences.

Gert [21] argued that it should be possible to justifiably violate moral norms. He classified norm violations into three categories: strongly justifiable, unjustifiable, and weakly justifiable. What counts as a strongly justifiable, unjustifiable, or weakly justifiable violation depends on whether all, some, or no impartial agents would publicly allow other violations [22]. Gert devised a two-step procedure to evaluate whether a violation is justifiable. In the first step, all the morally relevant information is ascertained. In the second step, one should consider the consequences of other people being able to violate the moral rule in similar circumstances. Considering our four types of examples, we can say that violations in Examples 3 and 4 are strongly justifiable, whereas the violations in Examples 1, 2 and 6 are unjustifiable.

Civil disobedience. The violation in Example 5 falls under the scope of civil disobedience [16]. With respect to civil disobedience, Gert states:

> Civil disobedience usually occurs only when a person believes that the law is unnecessarily causing significant evil. It is only justified when a person has some reason to believe that disobeying the law will do something toward lessening that evil. [21], p.204.

[3] both morally permissible or both morally impermissible

He also uses civil disobedience as an example of norm violations that can be punishable. Gert finds it justifiable to violate one norm in order to uphold another.

We should consider the literature on civil disobedience in digital environments. Klang [26] argues that civil disobedience should be tolerated online for the same reasons it is tolerated in the real world. To justify his claim, Klang reinterprets the criteria for disobedience for application in the digital environment. Specifically, he looks at: disobedience, civil, non-violence, and justification. Disobedience is grounded in the intention to protest a law that is in conflict with "more stringent obligations" [32]. The civil criterium is necessary to publish the disobedience. Non-violence is not a requirement, but a preference for civil disobedience because the use of violence in civil disobedience "has been shown to remove the focus of the message of protest and create a lack of sympathy for those who use it"[26]. Civil disobedience must be justifiable in the conflict of law with a specific moral principle. The classic justification of civil disobedience lies in a conflict of law with moral principle. Further criteria can be found in the acceptance of punishment [36] and the narrowness of the act: the civil dissident is looking to only violate those norms that they object to on moral grounds [19].

We observe that all of the examples have some characteristics which are different. These are: the attitudes of the agents to the publicity of their violation, the attitudes they have towards the sanctions that may follow, the expectations they have from the institutions reaction to their violations and the reasons they provide for their violation.

Attitudes to publicity and sanctions. In the context of our analysis, an important discerning property of a violation is how visible the violation has been. What we are actually considering here are two things: announcements of the violation by the agents themselves and the availability of witnesses to the violation. In the sense of the publicity of the violation we have three types of agents: those who want to hide the violation (Examples 1 and 2), those who want to advertise it (Examples 5 and 6), and the ambivalent agents (Example 3).

In the literature, there are multiple considerations surrounding the agent's visibility in conducting activities and the agent's public declarations about what they do and what they know.

Balbiani et al. [1] and Schwarzentruber [35] present the logic of agents looking at each other. They introduce the operator $\triangleright$ and use $a \triangleright b$ to mean 'a sees b'.

Dynamic epistemic logic [2] considers the specification of knowledge, public announcements, and common knowledge. We can see how the agents in Examples 3 and 4 would distinguish themselves by making public announcements "I violated the norm". There is a clear difference between agents who want to hide their violation and those for whom publicity of the violation does not play a role. Those who want to hide would avoid making public announcements from which it can be deduced that a norm has been violated (Examples 1 and 2). Those who do not want to hide do not need to be careful about their public announcements (Example 3).

In this paper we focus on the reasoning mechanisms of disobedience monitoring institutions and disobedient agents. to accomplish this present goal we do not need to specify the publicity of the violation in greater detail than: has the agent declared it and is there a probability that the agent is observed committing it.

Expectations. In our examples, we explicitly consider expectations. First, there is an expectation that the agents will follow the MAS norms. Additionally, in the same vein, there is a systemic expectation that the agents are informed that the institution expects that the agents are informed of what the norms are and are able to decide in which situations they apply.

Castelfranchi [10] considers the concept of expectation in the context of agent behaviour. He argues that expectations can be expressed as beliefs and goals as follows. An agent i expects p when they at time t they believe that p will be the case at time t'. Furthermore, an agent i has, at time t, the goal of knowing whether p is true or not at time t', for some $t \geq t'$. Castelfranchi further claims that expectations can be as follows. Positive expectation is when an agent is believing p and having the goal to confirm p. Negative expectation is when the agent is believing p and having the goal to confirm $\neg p$). Neutral expectations: believing p and having neither the goal to confirm p nor confirm $\neg p$. Ambivalent expectations: believing p and having both the goal of confirming p and the goal of confirming $\neg p$.

Bicchieri [4] considers the role expectations have in the context of social norms. In terms of the nature of expectations, she specifically states: "Social norms often engender expectations of compliance that are felt to be legitimate and close in a sense to 'having a right' to expect certain behaviours on the part of others, who, therefore, are perceived as 'having an obligation' to act in specific ways. This is because we have an ingrained tendency to move from what is to what ought to be and conclude that 'what is' must be right or good. Yet, apart from our longstanding habits of performing and expecting others to perform certain actions, there is no deeper foundation to these presumed 'rights and obligations', however intensely felt they might be. "

We want to design a compliance architecture and agent reasoning mechanism that can be applied to agentic AI. For this purpose, it is not necessary, and could be prohibitive, to claim a formalising language that is so expressive as to specify the goals and beliefs of individual agents and the analysis of Castelfranchi [10] is perhaps more detailed than what we need. We do take the same attitude as Bicchieri [4] that expectations do not presume rights and obligations.

Arguments for violation. We assume that the agents are able to provide reasons for the norm violation they have committed. Not all reasons have the same nature or serve the same purpose.

Norms are instituted for a reason or to serve a purpose. Consider, for example, the speed limit imposed on public roads. The purpose of this traffic regulation is to make the roads safer. As Bicchieri [4] argues, the institutional expectation that agents will comply with the norms is itself not sufficient for compliance. Norms, specifically legal and social norms, have sanctions. The sanction is there to further motivate compliance with the norm. If the sanction is too weak, the norm will be ignored. If the sanction is inadequately strong, consider executing people for walking on the grass when they should not, then the norm will disturb the efficiency of the society it is meant to help. Let us now consider the different arguments.

The ability or willingness of the agent to provide a justification, an explanation, or an excuse for their violation causes different treatment by the monitoring institution. An explanation is an argument that clarifies how the disobedience came about. Justifications are the reasons for which the agent should not be sanctioned for norm violation. Excuses are arguments that clarify why the disobedience was unavoidable, but in the sense that it was out of the agent's control to obey the norm.

Explanations of behaviour are the object of study in [28, 29] and attribution theory in psychology[4]. Malle distinguishes between intentional and unintentional behaviour. Reason explanations are what people use to explain the reasons why an agent had to act in a way it did (with intention). Cause explanations are "people's explanations of an unintentional behaviour that cite the causes that brought about the behaviour" [28].

An explanation, intuitively, is a statement or an account that aims to make something clear. Explanations are intended to transfer knowledge from automated systems to help people understand why decisions are made [13, 27]. The distinction between justifications and explanations has also been debated [25].

We argue that the term *explanations* is most adequate to name the arguments that support civil disobedience. Agents who are disobedient in this way clearly have the choice to not violate the norm. Their explanation is not an excuse because they intentionally violate the norm. The arguments that justify intentional norm violation with the purpose of avoiding sanctions we call *justifications.* This is relevant for Examples 3 and 4 who Type 2 need to demonstrate that the option of obeying the violated norm was not feasible within the existing normative system. Justifications must be grounded in some aspect of the decision-making process that led to the decision [18].

The agent may be asked to provide arguments why they unintentionally violated a norm. Thus is institutionally important information. Gert [21] recognises four types of excuses: epistemic limitations, duress, time limitations, and institutional pressures. Excuses are explicitly considered in Boella et al. [5] where five different types of excuse are distinguished. Epistemic excuses are related to the knowledge of the agents. Power-based excuses are based on lack of ability of the agent. Norm-based excuses are based on prioritisation among conflicting norms. Counts-as excuses are based on the interpretation of what does and does not apply. Sometimes this type of issue is called "normative uncertainty" in the literature. Lastly, social-based excuses are statements grounded in what is perceived as common or repeated behaviour in others: 'everyone does it'. In the scope of this paper we choose not do distinguish between types of excuses, but we consider it important to point out that there is a rich granularity in the literature that can be exploited in future work.

TYPES OF DISOBEDIENCE.

If we now revisit each of our examples, we can observe that there are four types of disobedience that can be discerned.

- **Type 1: Intentional violation.** Examples 1 and 2. The agent does not want to be observed violating the norm. They expect to get away without a sanction or that the sanction is worth it compared to other gains. The agent can provide excuses or explanations, but not justifications. The normative system should not change. The norm monitoring may need to be changed.
- **Type 2: Norm exception.** Examples 3 and 4. The agent does not care whether they are observed violating the norm. Can justify its actions. The agent expects not to be sanctioned (expects that the institution decides that the norm does not apply). The normative system does not change.
- **Type 3: Civil disobedience.** Example 5. The agent gives explanations for the act grounded in the need to change the normative system (and why?). The agent wants to be observed violating the norm and welcomes being sanctioned.
- **Type 4: Troll.** Example 6. The agent can provide no explanations, excuses, or justifications for their violation. The agent wants to be observed violating the norm. The agent wants to be sanctioned, that is the whole point of the violation. The normative system, specifically the sanctions, needs to change.

[4]Attribution theory is defined as the study of perceived causes of success and failure.

We argue that each type of violation supports a different system function: audit (Type 1), safety valve (Type 2), reform signal (Type 3), and noise filter (Type 4); dropping any one leads to misclassification and governance failure.

	Type 1	Type 2	Type 3	Type 4	Violation
Argument	-	j	e	-	x
Disclosure	h	i	v	v	i
Sanction	n	n	a	a	n
Expectation	-	nd	rc	sc	-

Table 1: The different properties of nonintentional and intentional violations (disobedience) along four dimensions.

Table 1 summarises our analysis. In the last column, we consider unintentional violations, whereas the other 4 types/columns correspond to disobedience. In the row 'Argument', the possible options are: an explanation (e), a justification (j), an excuse(x), or nothing (-). Lastly, the unintentional norm violation is also relevant to be correctly identified.

In the row 'Publicity' e denote the possible stance an agent can have towards disclosing, by themselves or from others, the fact they have intentionally violated a norm: hidden (h), visible (v), and indifferent (i). In the row 'Sanction', the possible options are: accept (a) or want to avoid (n). In the row 'Expectation', the possible options are: no detachment (nd), rule change(rc), sanction change (sc) and nothing (-).

COMPLIANCE ARCHITECTURE

We first consider the question of how to design an architecture for an institution that monitors compliance with norms. We organise compliance as a *monitor–classify–respond* pipeline over observed episodes. The pipeline is intentionally simple in its topography and strict in its control guards. At entry, a monitoring component observes the behaviour of agents and registers **episodes**. Each episode is turned into a *case.* Each case is end-to-end owned by a case manager to ensure traceability, service levels, and auditability. In the middle tier, a classifier assigns exactly one type to every case under the default setting. The classifier is implemented as a compact decision table. The final tier executes a type-specific response workflow. Our position is functional. Rather than the labels themselves, we emphasise the *roles* that different kinds of disobedience play in social systems (deterrence, safety valve, reform signalling, and noise filter).

Episodes, publicity, and reasons

We define an episode as a record $e = \langle id, actor, act, time, ctx \rangle$ augmented with two visibility *attributes*: a *disclosure stance* $\mathrm{Disc}(e)$ and an *observability score* $\mathrm{Obs}(e)$. We write $\mathrm{Disc}(e) \in \{hide, partial, public\}$ for the agent's disclosure stance and $\mathrm{Obs}(e) \in [0, 1]$ for the witnessing level derived from sensors, logs, or third-party reports. *Publicity* is a boolean value determined by

$$\mathrm{Public}(e) \equiv \big[\mathrm{Disc}(e) = public\big] \;\vee\; \big[\mathrm{Obs}(e) \geq \theta_{\mathrm{pub}}\big].$$

$\mathrm{Obs}(e)$ is computed from auditable sources (e.g., device logs, platform telemetry, third–party reports); the threshold θ_{pub} is a documented policy parameter calibrated for each deployment. Borderline cases (e.g., conflicting reports) are routed to the manual triage noted below; the triage amends the episode fields (with rationale) or confirms them for classification.

Episodes also carry a *reason tag* $\mathrm{reason}(e) \in \{none,\ justification,\ explanation,\ excuse\}$. We use *justification* for case-specific reasons that aim to avoid sanction; *explanation* for systemic reasons that point to rule change; and *excuse* for unintentional breaches (e.g., epistemic limitation, duress, time pressure).

Example 7. The observables in Sections 1–2 fill these fields directly, listed in Table 2.

e	Description	$\mathrm{Disc}(e)$	$\mathrm{reason}(e)$
Example 1	speeding for convenience	hide	none
Example 2	robot detour to charge	hide	none
Example 3	medical emergency	partial	justification
Example 4	guide-dog safety	partial	justification
Example 5	social network ban	public	explanation
Example 6	boastful speeding	public	none

Table 2: Episodes and their disclosure and reason values.

Every episode produces a *case* with a unique identifier. Every episode produces a case with a unique identifier. Cases are recorded before any decision and are never processed off record. The case backbone prevents two practical failures that are common in large organisations: items that fall between proverbial chairs and items that are worked twice by different offices. A case cannot leave a queue without a recorded status change, and every status must have a defined successor.

Tier 1: Norm monitor (detachment and exceptions)

For the deontic statement of a norm to be upheld, an institution first must decide to detach the norm. For example, consider a parent that says: If you are not quiet, you are not allowed to have ice cream. If the child is not quiet, the parent may still decide to allow the ice cream. We say: the parent did not detach the norm. In our compliance architecture, the norm monitor evaluates whether any norm is *detached* in the episode context and whether an exception defeats detachment. We write

$$\mathrm{NM}(e) \in \{compliant,\ violated,\ justified\_exception\},$$

computed from $\mathrm{Detach}(N, C_e)$ together with the exception list $E(n)$. If an exception defeats detachment, there is no violation. This first tier ensures that type classification is only applied to *operative* norms and that validated exceptions do not enter the violation logic at all.

Example 8. Examples 3 and 4 are resolved here as justified exception: the norm does not detach in light of emergency safety considerations (Example 3) or an unsafe command (Example 4). Examples 1-2 and Examples 5-6 remain *violated* and move to typing in Tier 2.

Tier 2: Type classifier (visibility and reasons)

In this next stage of processing, the classifier consumes $(e, \mathrm{NM}(e), \mathrm{Public}(e), \mathrm{reason}(e))$ and outputs *exactly one* type

$$t \in \mathrm{Type} = \{Obedient,\ Unintentional,\ Violation,\ Exception,\ Civil,\ Troll\}.$$

Two control guards are enforced: *completeness* (every case receives a type) and *exclusivity* (at most one type per case). When a guard would fail, the case is sent to a short manual triage that amends the episode fields or confirms a single type with rationale.

The decision table is compact. If $\mathrm{NM}(e) = compliant$ the output is *Obedient*; if $\mathrm{NM}(e) = justified\_exception$ the output is *Exception*. Otherwise $\mathrm{NM}(e) = violated$ and we look at visibility and reasons:

- if $\mathrm{Public}(e)$ and $\mathrm{reason}(e) = explanation$ and $\textsc{SanctionReady}(e)$ and $\textsc{ProtestDeclared}(e)$, then *Civil*;
- else if $\mathrm{Public}(e)$ and $\mathrm{reason}(e) = none$, then *Troll*;
- else if $\mathrm{reason}(e) = excuse$, then *Unintentional*;
- else *Violation*.

Example 9. With NM = *violated*: Examples 1-2 are typed as *Violation* (hidden, no reason); Example 5 is *Civil* (public, systemic explanation, sanction-ready, protest declared); Example 6 is *Troll* (public, no reason).

The exclusivity setting that we enforce here reflects the characteristics of a *classifier*. We could also have designed a variant *filter* – where several predicates may hold simultaneously. At present we consider this option only as a possible future extension; precedence rules would then be applied at the dispatch. We keep the default classifier for clarity and predictable routing.

Tier 3: Response management (type-specific workflows)

In the next stage of our pipeline, each type is routed to a distinct workflow. Cases of type *Violation* enter a deterrence lane where the sanction is issued and settled. Cases of type *Unintentional* are analysed for system improvement while proportionate sanctions may still apply. Cases of type *Exception* are resolved in an exception lane that produces a waiver with an explanation; the attched rationale is archived and later mined for adjustments to the detachment criteria. Cases of type *Civil* proceed in two branches: the sanction is applied to preserve procedural fairness, while a protest record is

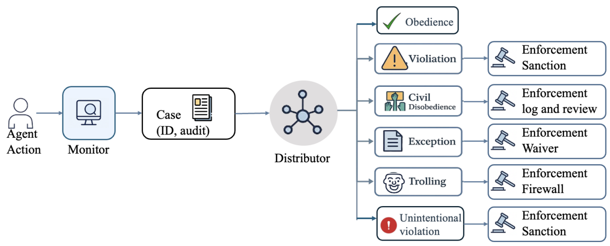

Figure 1: Normative routing from observed action to response. The monitor validates detachment and exceptions; the classifier assigns exactly one type; each lane executes a distinct workflow.

created and forwarded to a review board that considers whether the underlying rule needs revision. Cases of type *Troll* trigger sanction and a *noise filter* so that disruptive behaviour does not influence policy review. Segregation of duties is observed, and the case backbone enforces end-to-end visibility so that each workflow closes with a clear terminal state.

Example 10. Back to our running illustration. Examples 1–2 route to *Violation* → sanction (deterrence/audit). Examples 3–4 route to *Exception* → waiver with explanation (safety valve). Example 5 routes to *Civil* → sanction, protest log, review (reform signal). Example 6 routes to *Troll* → sanction plus noise filter (excluded from reform). See also Figure 1.

Algorithmic view

We now present the per-episode algorithm, Algorithm 1, that follows the three tiers we have described. The algorithm makes explicit the order of checks (detachment before typing) and aligns the branches with the response lanes.

Governance and feedback

The enforcement tier is not a cul-de-sac. Exception rationales are periodically reviewed to codify recurring patterns as detachment criteria for the corresponding norms. Civil records are summarised and sent to a governance calendar where proposals for revision are evaluated with their supporting evidence. Metrics on unintentional cases inform transparency and education efforts. Trolling metrics calibrate noise filter without entering the reform channel. This feedback is light-weight to keep the pipeline operational, yet it is sufficient to connect day-to-day processing with policy upkeep. Two properties make the architecture dependable in practice: the *coverage guard* (every observed episode becomes a case and every case leaves Tier 2 with a type) and the *exclusivity guard* (under the default classifier, exactly one lane is taken; an optional filter variant resolves conflicts at dispatch). Together with persistent case ownership, these properties prevent the real administrative pathologies of lost items, duplicate processing, and contradictory outcomes.

Algorithm 1: Per-episode processing (monitor → classify → respond)

Require: Episode e with Disc, Obs, reason; norms N; threshold θ_{pub}

$(\textit{detached}, \textit{exc}) \leftarrow \textsc{DetachAndExceptions}(N, e)$
if $\neg\textit{detached}$ **then**
  return *Obedient*
end if
if *exc* **then**
  return *Exception*
end if
$v \leftarrow (\text{Disc}(e) = \textit{public}) \lor (\text{Obs}(e) \geq \theta_{\text{pub}})$
if $v \land (\text{reason}(e) = \textit{explanation}) \land \textsc{SanctionReady}(e) \land \textsc{ProtestDeclared}(e)$ **then**
  $t \leftarrow \textit{Civil}$
else if $v \land (\text{reason}(e) = \textit{none})$ **then**
  $t \leftarrow \textit{Troll}$
else if $\text{reason}(e) = \textit{excuse}$ **then**
  $t \leftarrow \textit{Unintentional}$
else
  $t \leftarrow \textit{Violation}$
end if
$\textsc{Dispatch}(t)$ ▷ *Exception*: waive; *Civil*: sanction+log+review; *Troll*: sanction+noise_filter; *Unintentional*: proportionate response+system improvement; *Violation*: sanction
return t

AGENT DECISION MAKING: REASON–BASED DISOBEDIENCE

The previous section fixed the institutional side: episodes are monitored, typed, and routed to one of the response lanes. We now turn to the *agent*'s side and explain how a single course of action is selected in light of that framework. We adopt Tucker's *dual-scale theory of reasons* as the decision procedure [37]. Dual-scale separates what makes an option *permissible* from what makes it *required*. Dual-scale treats reasons as *contrastive*: a consideration counts for act a only relative to an alternative b, and its force may differ across contrasts.

Tucker [37] distinguishes between three weights: *justifying weight* (JW), which pushes an act toward permissibility; *requiring weight* (RW), which pushes alternatives toward impermissibility; and *commending weight* (CW), which orders acts that are already permissible without altering their deontic status. The Permission Scale asks whether $\text{JW}(a \parallel b)$ meets or exceeds $\text{RW}(b \parallel a)$; the Commitment Scale mirrors it by flipping sides; an act becomes *required* precisely when it is permissible against every rival and each rival is impermissible against it. This approach preserves the difference between "may," "must," and "best among what may be done". A single blended score would have blured this distinction.

We adopt Tucker's dual-scale theory of reasons as a *general* decision procedure over contrastive reasons. Concretely, we abstract the theory to a logic of pairwise comparisons: each ground (g, a, b) contributes two kinds of weight—one *in favour of* a and one *against* b—originally read as *justifying* and *requiring* weight in Tucker's theory [37], but here usable under pro/con reading suitable to the decision task. This makes the agent reasoning mechanism we propose applicable beyond pure permissibility tests: whenever alternatives have pros and cons, we can evaluate them contrastively and then choose among the alternatives that survive pairwise competition.

Because we are looking at alternatives that survive pairwise competition, what we have is a tournament over the options A. We have the following choice set that matches the behaviours observable by the institution:

$$A = \{\textsc{Obey}, \textsc{Hidden}, \textsc{Exception}, \textsc{Civil}\}.$$

Troll is not introduced as a deliberative option; it is an institutional classification that results when a public violation is declared without reasons. Each option in A implies a different signalling profile for the episode: Obey performs the compliant act; Hidden attempts a non-public breach and withholds reasons; Exception provides a case-specific justification whose aim is to avoid sanction; Civil acts publicly with a systemic explanation, declares sanction-readiness, and issues a protest statement. These observables are exactly those read by the classifier in the previous section.

Formal rules (disobedience). For any ordered pair (a, b) of distinct options, let $\mathcal{G}(a \parallel b)$ be the contrast-indexed set of grounds supporting a against b. Each $g \in \mathcal{G}(a \parallel b)$ contributes one or more reasons with weight $w_g \in \mathbb{N}$:

- $\mathrm{JW}(a \parallel b) = \sum_{g \in \mathcal{G}^{\mathrm{just}}(a \| b)} w_g$
- $\mathrm{RW}(a \parallel b) = \sum_{g \in \mathcal{G}^{\mathrm{req}}(a \| b)} w_g$
- $\mathrm{CW}(a) = \sum_{g \in \mathcal{G}^{\mathrm{com}}(a)} w_g$

Dual-scale verdicts are pairwise:

- a is *permissible* against b iff $\mathrm{JW}(a \parallel b) \geq \mathrm{RW}(b \parallel a)$;
- a is a *commitment* against b iff $\mathrm{RW}(a \parallel b) > \mathrm{JW}(b \parallel a)$;
- a is *required* when both hold against every rival.

With multiple options, this yields a *tournament*: an action is chosen *iff* it wins all pairwise contests. If several options remain permissible without a Condorcet winner (an option that defeats all others in a pairwise comparison), CW ranks within the permissible set; if no option is permissible, the default is Obey.

We use the following symbols in the quantitative reasoning below. $\hat{p}_{\mathrm{det}}(a) \in [0, 1]$ denotes the agent's estimated probability that act a will be detected by the monitoring system. $\hat{s}(a) \geq 0$ denotes the expected sanction severity (e.g., monetary fine or reputational cost) conditional on detection of a. $\hat{v}_{\mathrm{reform}} \geq 0$ denotes the expected institutional value that a civil act may contribute to systemic reform when publicly registered. $\mathrm{cred}(e) \in [0, 1]$ denotes the credibility of the episode e, reflecting the quality of its systemic explanation together with the agent's sanction-readiness and protest declaration. All of these estimates are local to the deliberating agent and serve as parameters in the expectation calculations below.

Quantitative expectations as reasons. Two environment-facing expectations are folded into the scales as reasons with weights rather than as external utilities. *Expected penalty* for a is

$$E[\text{Penalty} \mid a] = \hat{p}_{\mathrm{det}}(a) \cdot \hat{s}(a);$$

this contributes to the *requiring* side *against* the more exposed act in each contrast (e.g., increases $\mathrm{RW}(\textsc{Hidden} \parallel \textsc{Obey})$ when concealment is weak, or $\mathrm{RW}(\textsc{Civil} \parallel \textsc{Obey})$ when penalties are stiff). *Expected reform value* is relevant to Civil:

$$E[\text{Reform} \mid \textsc{Civil}] = \hat{v}_{\mathrm{reform}} \cdot \mathrm{cred}(e),$$

where $\mathrm{cred}(e)$ reflects the quality of the systemic explanation together with sanction-readiness. This contributes *commending* weight to Civil and, where policy permits, may also contribute justifying weight.

The option set is pruned before the tournament. Exception is available only with a plausible justification plan that the institution could validate. Civil is available only if publicity is feasible and the agent can both declare sanction-readiness and issue a protest. These guards keep the choice set aligned with what the surrounding framework can recognise.

Example 11. Agent decision on civil posting on a platform In this scenario there is no recognised educational exception, so the available options are $\{\textsc{Obey}, \textsc{Hidden}, \textsc{Civil}\}$. We present three contrasts in the same format as the rights, see Tables 3, 4 and 5.

Table 3: Contrast 1: CIVIL vs. OBEY

$(g, a \parallel b)$	JW	RW	Comment
(public_health, civil || obey)	3	2	early detection / breastfeeding benefits
(equality_legitimacy, civil || obey)	2	1	rule impacts groups unevenly
(reform_visibility, civil || obey)	2	1	public act needed to signal reform
(credibility_sanction_ready, civil || obey)	1	0	acceptance of penalty lends credibility
(policy_compliance, obey || civil)	2	1	follow platform terms
(predictability, obey || civil)	1	2	uniform enforcement

Totals: $\mathrm{JW}(\textsc{civil} \parallel \textsc{obey}) = 8 \geq \mathrm{RW}(\textsc{obey} \parallel \textsc{civil}) = 3$, so civil is permissible; $\mathrm{RW}(\textsc{civil} \parallel \textsc{obey}) = 4 > \mathrm{JW}(\textsc{obey} \parallel \textsc{civil}) = 3$, so civil is a commitment. Thus, against obey, civil is required.

Table 4: Contrast 2: CIVIL vs. HIDDEN

$(g, a \parallel b)$	JW	RW	Comment
(reform_visibility, civil || hidden)	3	2	hidden posts cannot signal reform
(honesty_legitimacy, civil || hidden)	2	1	open stance avoids misrepresentation
(credibility_sanction_ready, civil || hidden)	1	0	willingness to bear costs
(sanction_avoidance, hidden || civil)	1	2	concealment to keep account
(lower_detection_risk, hidden || civil)	1	2	reduced exposure

Totals: $\mathrm{JW}(\textsc{civil} \parallel \textsc{hidden}) = 6 \geq \mathrm{RW}(\textsc{hidden} \parallel \textsc{civil}) = 4$, so civil is permissible; $\mathrm{RW}(\textsc{civil} \parallel \textsc{hidden}) = 3 > \mathrm{JW}(\textsc{hidden} \parallel \textsc{civil}) = 2$, so civil is a commitment. Thus, against hidden, civil is required.

Table 5: Contrast 3: HIDDEN vs. OBEY

$(g, a \parallel b)$	JW	RW	Comment
(reach_quickly, hidden || obey)	1	2	faster sharing within a small circle
(self_interest, hidden || obey)	1	1	convenience to the poster
(policy_compliance, obey || hidden)	2	1	terms and fair procedure
(transparency, obey || hidden)	1	2	avoid covert breaches

Totals: $\mathrm{JW}(\textsc{hidden} \parallel \textsc{obey}) = 2 < \mathrm{RW}(\textsc{obey} \parallel \textsc{hidden}) = 3$, so hidden is not permissible; $\mathrm{RW}(\textsc{obey} \parallel \textsc{hidden}) = 3 > \mathrm{JW}(\textsc{hidden} \parallel \textsc{obey}) = 2$, so obey is a commitment against hidden.

After the guards and contrasts are applied, the tournament selects the action that wins all pairwise contests. In this scenario, Civil defeats both Obey and Hidden on the permission and commitment scales, and is therefore chosen. If several options remained permissible without a winner, commending weight would rank among them; if none were permissible, Obey would be taken as the residual choice. The resulting observables are then emitted (publicity, reason type, sanction-readiness, protest declaration) so the institutional classifier in the previous section reads the episode as intended.

CONCLUSION AND FUTURE WORK

In this article, we present a base architecture for reasoning with and about disobedience in multi-agent systems. We distinguish four types of intentional norm violation, i.e., disobedience from unintentional violations. To the best of our knowledge, this topic has not been extensively considered in normative multi-agent systems, the field typically concerned with norm management. We argue the importance of normative MAS researching and engineering the subject of agents, which we intentionally violate norms. To that end, this paper should serve as a first contribution towards a rich prospect of future work, which we outline in this section.

Having intelligent multi-agent systems is becoming particularly important with the increased capability of Agentic AI is presenting a different approach to accomplishing artificial intelligence agents [7]. The smooth operation of a MAS requires norms for governing the operation in a shared environment. Multi-agent systems design typically includes not only constructing the agent itself but also the agent environment, allowing for institutions to be implemented as part of the same environment. Agentic AI are not agents that are constructed following the same architecture or paradigm, in contrast to, say, BDI agents [24]. But we still need to enable them to reason about norms. We also need to build digital environments that can handle intelligent agent behaviour in the correct way. Failure to do so makes for ineffective governing of multi-agent systems, as well as disrupting societal instruments such as civil disobedience.

What we present in this article is the basic construction of a compliance architecture. In this construction, many details of the agent and institutional behaviour are abstracted. Enriching each of these details is a promising direction for future work.

First, we assume that the agents are truthful, which is a relevant assumption when we consider their reasons for violating the norm. An agent who intentionally violates a norm for their own gain can find it beneficial to offer an excuse to pass off the disobedience as an unintentional violation to perhaps profit from it. The strategic behaviour of the disobedient agent should be considered. But what also needs to be considered is the hegemonic power of the institution.

In our current pipeline, once the violation, or event, is typed, the institution proceeds with its process. We implicitly assume that the institution is infallible. In practice, agents have the right to contest decisions about detachment and sanctions [17]. Future work should explore the contestation process as part of the compliance architecture.

In order to further specify the institutional treatment of violators and that includes the option to contest the institutional decisions, the structure of the reasons needs to be made more expressive. The justifications offered must be grounded in the reasons why the norm was instituted in the first place. For example, the justification for why one speeds to the hospital can be found in the reason for the prohibition for speeding, which is safety. In the context of a medical emergency, the passenger is less safe when the vehicle respects the speed limit. However, even ambulances that are allowed to drive through red lights and speed limits are not allowed to drive recklessly. The case of civil disobedience offers a particularly rich set of challenges. The reason for civil disobedience is, in general, the conflict between legal and moral norms. One should not engage in norm violation for personal gain and explain it away as civil disobedience. Institutionally, civil disobedience and trolling differ only in the willingness of the agent to provide an explanation. Having the ability to judge the legitimacy of the explanation is an important capability of institutions that merits further study. Lastly, excuses can also be misused. We do not distinguish between different types of excuses, but some can be easier to institutionally verify than others. In summary, verifying the reasons for the agents is a necessary future work direction.

As we discussed in the related work section, the publicity, or observability, of a norm violation plays an important role in its institutional handling. In our architecture visibility is reduced to a Boolean signal. Although we distinguish three types of disclosure that an agent can make, we strongly simplify the observability of an act to a probability value: $\mathrm{Obs}(e) \in [0, 1]$ for the witnessing level derived from sensors, logs, or third-party reports. It is clear that the topic of evidence is material evidence and witness reports of different relevance and reliability. The likelihood of being observed committing a violation is not the only reason why some agents will or will not violate a norm. Some people wait at a red light on a pedestrian crossing when there are no cameras, no witnesses, and no traffic. Others cross when all three are present. The attitude of the agent towards moral and legal norms is part of the agent type and subject to different agent designs. All of our agents are of the same type. Exploring a true multi-agent system of multiple agent types and perhaps institutions with different jurisdictions are a particularly hard challenge for future work.

In our types of disobedience, we did not include the cases of whistleblowing. Whistleblowing is a violation of privacy and sometimes safety norms. It happens when an agent discloses information from a public or private organisation to reveal issues of immediate or potential danger to the public [34]. Although whistleblowing and civil disobedience are similar, they are not the same. Unlike civil disobedience, whistleblowing can sometimes be done anonymously. In our present design, we chose not to model whistleblowing because it is specific to privacy, which would detract from the general compliance architecture we propose.

Lastly, an implementation of a prototype of a MAS in which disobedience happens and it is monitors is necessary to further validate our designs, and to also uncover possible practical points of failure that need to be addressed.

REFERENCES

[1] Philippe Balbiani, Olivier Gasquet, and FranÃ§ois Schwarzentruber. 2012. Agents that look at one another. *Logic Journal of the IGPL* 21, 3 (12 2012), 438–467. https://doi.org/10.1093/jigpal/jzs052 arXiv:https://academic.oup.com/jigpal/article-pdf/21/3/438/1850157/jzs052.pdf
[2] Alexandru Baltag and Bryan Renne. 2016. Dynamic Epistemic Logic. In *The Stanford Encyclopedia of Philosophy* (Winter 2016 ed.), Edward N. Zalta (Ed.).

Metaphysics Research Lab, Stanford University.

[3] Trevor Bench-Capon and Sanjay Modgil. 2016. When and How to Violate Norms. *Frontiers in Artificial Intelligence and Applications* 294:Legal Knowledge and Information Systems (2016), 43–52. https://www.csc.liv.ac.uk/~tbc/publications/Bench-Capon_15.pdf.

[4] Cristina Bicchieri. 2014. Norms, Conventions, and the Power of Expectations. In *Philosophy of Social Science: A New Introduction*, Nancy Cartwright and Eleonora Montuschi (Eds.). Oxford University Press.

[5] Guido Boella, Jan Broersen, Leendert van der Torre, and Serena Villata. 2009. Representing Excuses in Social Dependence Networks. In *AI*IA 2009: Emergent Perspectives in Artificial Intelligence*, Roberto Serra and Rita Cucchiara (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 365–374.

[6] Guido Boella, Leendert Torre, and Harko Verhagen. 2006. Introduction to normative multiagent systems. *Comput. Math. Organ. Theory* 12, 2–3 (Oct. 2006), 71–79. https://doi.org/10.1007/s10588-006-9537-7

[7] Vincent Botti. 2025. Agentic AI and Multiagentic: Are We Reinventing the Wheel? arXiv:2506.01463 [cs.MA] https://arxiv.org/abs/2506.01463

[8] Philip Brey and Brandt Dainow. 2024. Ethics by design for artificial intelligence. *AI and Ethics* 4, 4 (Nov. 2024), 1265–1277. https://doi.org/10.1007/s43681-023-00330-4

[9] Nils Bulling, Mehdi Dastani, and Max Knobbout. 2013. Monitoring norm violations in multi-agent systems. In *Proceedings of the 2013 international conference on Autonomous agents and multi-agent systems*. 491–498.

[10] Cristiano Castelfranchi. 2005. Mind as an Anticipatory Device: For a Theory of Expectations. In *Brain, Vision, and Artificial Intelligence*, Massimo De Gregorio, Vito Di Maio, Maria Frucci, and Carlo Musio (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 258–276.

[11] Amit K. Chopra, Leendert van der Torre, H. Verhagen, and Serena Villata. 2018. *Handbook of Normative Multiagent Systems*. College Publications. https://api.semanticscholar.org/CorpusID:203708095

[12] Philip R. Cohen and Hector J. Levesque. 1990. Intention is choice with commitment. *Artificial Intelligence* 42, 2 (1990), 213–261. https://doi.org/10.1016/0004-3702(90)90055-5

[13] European Commission, Content Directorate-General for Communications Networks, Technology, and Grupa ekspert w wysokiego szczebla ds. sztucznej inteligencji. 2019. *Ethics guidelines for trustworthy AI*. Publications Office. https://doi.org/doi/10.2759/346720

[14] Matteo Cristani and Elisa Burato. 2009. Approximate solutions of moral dilemmas in multiple agent system. *Knowledge and Information Systems* 18, 2 (Feb. 2009), 157–181. https://doi.org/10.1007/s10115-008-0172-0

[15] Mehdi Dastani, Paolo Torroni, and Neil Yorke-Smith. 2018. Monitoring norms: a multi-disciplinary perspective. *The Knowledge Engineering Review* 33 (2018), e25. https://doi.org/10.1017/S0269888918000267

[16] Candice Delmas and Kimberley Brownlee. 2024. Civil Disobedience. In *The Stanford Encyclopedia of Philosophy* (Fall 2024 ed.), Edward N. Zalta and Uri Nodelman (Eds.). Metaphysics Research Lab, Stanford University.

[17] Virginia Dignum, Matteo Baldoni, Cristina Baroglio, Maurizio Caon, Raja Chatila, Louise Dennis, Gonzalo Génova, Galit Haim, Malte S. Kließ, Maite Lopez-Sanchez, Roberto Micalizio, Juan Pavón, Marija Slavkovik, Matthijs Smakman, Marlies van Steenbergen, Stefano Tedeschi, Leon van der Toree, Serena Villata, and Tristan de Wildt. 2018. Ethics by Design: Necessity or Curse?. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society* (New Orleans, LA, USA) *(AIES '18)*. Association for Computing Machinery, New York, NY, USA, 60–66. https://doi.org/10.1145/3278721.3278745

[18] Virginia Dignum, Loizos Michael, Juan Carlos Nieves, Marija Slavkovik, Julliett Suarez, and Andreas Theodorou. 2025. Contesting Black-Box AI Decisions. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2025, Detroit, MI, USA, May 19-23, 2025*, Sanmay Das, Ann Nowé, and Yevgeniy Vorobeychik (Eds.). International Foundation for Autonomous Agents and Multiagent Systems / ACM, 2854–2858. https://doi.org/10.5555/3709347.3744034

[19] N. Ben Fairweather. 1999. The Future of Environmental Direct Action: A Case for Tolerating Disobedience. In *Environmental Futures*, N. Ben Fairweather, Sue Elworthy, Matt Stroh, and Piers H.G. Stroh Stephens (Eds.). Macmillan.

[20] Nicoletta Fornara, Henrique Lopes Cardoso, Pablo Noriega, Eugénio Oliveira, Charalampos Tampitsikas, and Michael I. Schumacher. 2013. *Modelling Agent Institutions*. Springer Netherlands, Dordrecht, 277–307. https://doi.org/10.1007/978-94-007-5583-3_18

[21] Bernard Gert. 1998. *Morality: Its Nature and Justification*. Oxford University Press, New York.

[22] Matthew W. Hallgarth. 2003. *Bernard Gert'S Theory of Moral Rules and American Professional Military Ethics*. Ph.D. Dissertation. University of Florida.

[23] Shawn He, Surangika Ranathunga, Stephen Cranefield, and Bastin Tony Roy Savarimuthu. 2025. Norm Violation Detection in a Multi-Agent Systems Using Large Language Models. In *Coordination, Organizations, Institutions, Norms, and Ethics for Governance of Multi-Agent Systems XVII*, Stephen Cranefield, Luis Gustavo Nardin, and Nathan Lloyd (Eds.). Springer Nature Switzerland, Cham, 146–159.

[24] Andreas Herzig, Emiliano Lorini, Laurent Perrussel, and Zhanhao Xiao. 2017. BDI Logics for BDI Architectures: Old Problems, New Perspectives. *Künstliche Intell.* 31, 1 (2017), 73–83. https://doi.org/10.1007/S13218-016-0457-5

[25] Tziporah Karachkoff. 1988. Explaining and Justifying. *Informal Logic* 10 (1988). https://doi.org/10.22329/il.v10i1.2635

[26] Mathias Klang. 2004. Civil disobedience online. *Journal of Information, Communication and Ethics in Society* 2, 2 (05 2004), 75–83. https://doi.org/10.1108/14779960480000244 arXiv:https://www.emerald.com/jices/article-pdf/2/2/75/1461104/14779960480000244.pdf

[27] Christian Lahusen, Martino Maggetti, and Marija Slavkovik. 2024. Trust, trustworthiness and AI governance. *Scientific Reports* 14, 1 (Sept. 2024), 20752. https://doi.org/10.1038/s41598-024-71761-0

[28] Bertram F. Malle. 1999. How People Explain Behavior: A New Theoretical Framework. *Personality and Social Psychology Review* 3, 1 (1999), 23–48. https://doi.org/10.1207/s15327957pspr0301_2 PMID: 15647146.

[29] Bertram F. Malle. 2011. Chapter six - Time to Give Up the Dogmas of Attribution: An Alternative Theory of Behavior Explanation. Advances in Experimental Social Psychology, Vol. 44. Academic Press, 297–352. https://doi.org/10.1016/B978-0-12-385522-0.00006-8

[30] Smitha Milli, Dylan Hadfield-Menell, Anca Dragan, and Stuart Russell. 2017. Should robots be obedient?. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence* (Melbourne, Australia) *(IJCAI'17)*. AAAI Press, 4754–4760.

[31] Reuth Mirsky. 2025. Artificial intelligent disobedience: Rethinking the agency of our artificial teammates. *AI Magazine* 46, 2 (2025), e70011. https://doi.org/10.1002/aaai.70011 arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/aaai.70011

[32] John Rawls. 1971. *A Theory of Justice: Original Edition*. Harvard University Press. http://www.jstor.org/stable/j.ctvjf9z6v

[33] Stuart J. Russell and Peter Norvig. 2020. *Artificial Intelligence: A Modern Approach (4th Edition)*. Pearson. http://aima.cs.berkeley.edu/

[34] Daniele Santoro and Manohar Kumar. 2017. A Justification of Whistleblowing. *Philosophy and Social Criticism* 43, 7 (2017), 669–684. https://doi.org/10.1177/0191453717708469

[35] François Schwarzentruber. 2011. Seeing, Knowledge and Common Knowledge. In *Logic, Rationality, and Interaction*, Hans van Ditmarsch, Jérôme Lang, and Shier Ju (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 258–271.

[36] Peter Singer. 1973. *Democracy and Disobedience*. Oxford University Press.

[37] Chris Tucker. 2025. *The Weight of Reasons: A Framework for Ethics*. Oxford University Press, New York.